LLM as a Clustering Oracle: Necessary at Every Stage?

Testing the robustness of LLM-guided text clustering before, during, and after partitioning

Wissal Zghrata · Rim Zeidan · Waël Akil

Université Paris Cité, Master Machine Learning pour la Science des Données, 2025/2026 · **Reference Paper:** *Large Language Models Enable Few-Shot Clustering*, Viswanathan et al. (2024) · arXiv:2307.00524

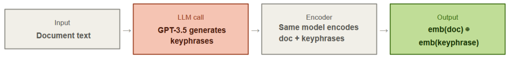
Project repository: wixii/llm-clustering-oracle-analysis

ENRICH *LLM intervention before clustering*

§ THE PAPER

`$1–10 / DATASET`

The LLM generates **task-relevant keyphrases**, encoded by the same model and **concatenated with the original embedding**.

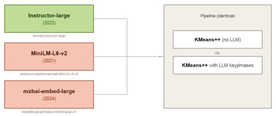


Gains on **3/5 datasets** · CLINC: **+5.8 acc** · Encoder = Instructor

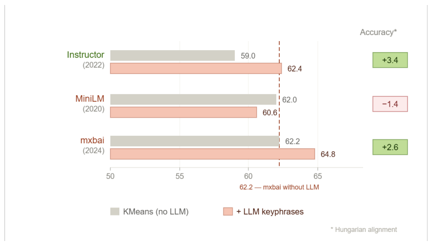
EVAL 1 — MODERN ENCODER

"Does the gain come from the LLM or from a dated encoder?"

We replace Instructor (2022) with mxbai-embed-large (2024), same pipeline.



KEY RESULT



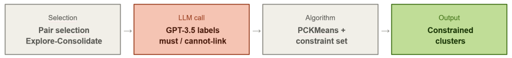
mxbai without LLM (62.2) ≈ Instructor with LLM (62.4)

CONSTRAIN *LLM intervention during clustering*

§ THE PAPER

`$10–42 / DATASET`

The LLM provides **must-link / cannot-link constraints** during iterative partitioning, acting as a **pairwise supervision signal**.

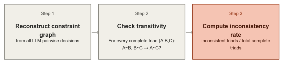


20,000 pairs/dataset · Most expensive stage

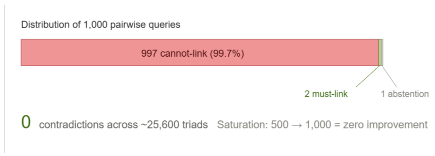
EVAL 2 — TRANSITIVE INCONSISTENCIES

"Are 20,000 LLM constraints logically coherent?"

We reconstruct constraint graphs from 1,000 queries (~12,800 complete triads per encoder).



KEY RESULT



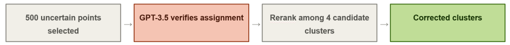
Trivially consistent: 99.7% cannot-link yields supervision without diversity.

CORRECT *LLM intervention after clustering*

§ THE PAPER

`$3–13 / DATASET`

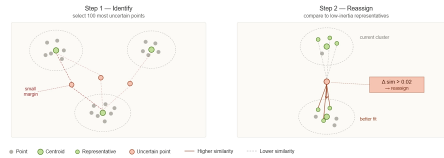
The LLM **post-hoc reassigns ambiguous points** to clusters whose label better matches their content.



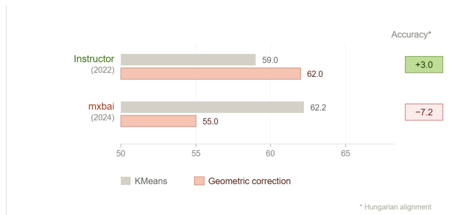
500 uncertain points · Gains: **+0.1 to +5.2**

EVAL 3 — GEOMETRIC BASELINE

"Is the LLM needed for correction or does simple geometry suffice?"



KEY RESULT



On the paper's encoder, geometry matches the LLM at zero cost. On a strong encoder, correction degrades.

Eval 4 — Generalization (OOD)

We apply the pairwise pseudo-oracle to **20 Newsgroups**, a domain and text style absent from the original benchmark, to test whether **LLM gains transfer out-of-distribution**.

+6.3
accuracy gain
75.2 → 81.6

40%
oracle abstention rate
pseudo-oracle declined to label

OVERALL INTERPRETATION

Transferability is **suggested but not established**. The pseudo-oracle helps (**+6.3 acc**), but **40% abstention** reveals degraded confidence outside intent-clustering. Broader generalization cannot be claimed from a 300-document pilot.

The framework remains valuable as an analytical lens but the gains are contingent on encoder, constraint distribution, and initial clustering quality.

THREE PROMISES, THREE VERDICTS

- `Enrich` **Encoder-dependent, not absolute. A 2024 encoder without LLM matches Instructor with LLM.**
- `Constrain` **Trivially consistent but informationally poor. Volume without quality.**
- `Correct` **Geometry suffices on the paper's encoder. On a strong encoder, both corrections degrade.**

REFERENCES

[1] Viswanathan et al. (2024). Large Language Models Enable Few-Shot Clustering. TACL.
[2] Basu et al. (2004). Active Semi-Supervision for Pairwise Constrained Clustering. SDM.
[3] Su et al. (2022). One Embedder, Any Task: Instruction-Finetuned Text Embeddings.
[4] Li et al. (2024). Towards General Text Embeddings with Multi-Stage Contrastive Learning.
[5] Lin et al. (2025). ClusterLLM. EMNLP.
[6] Reimers & Gurevych (2019). Sentence-BERT. EMNLP.